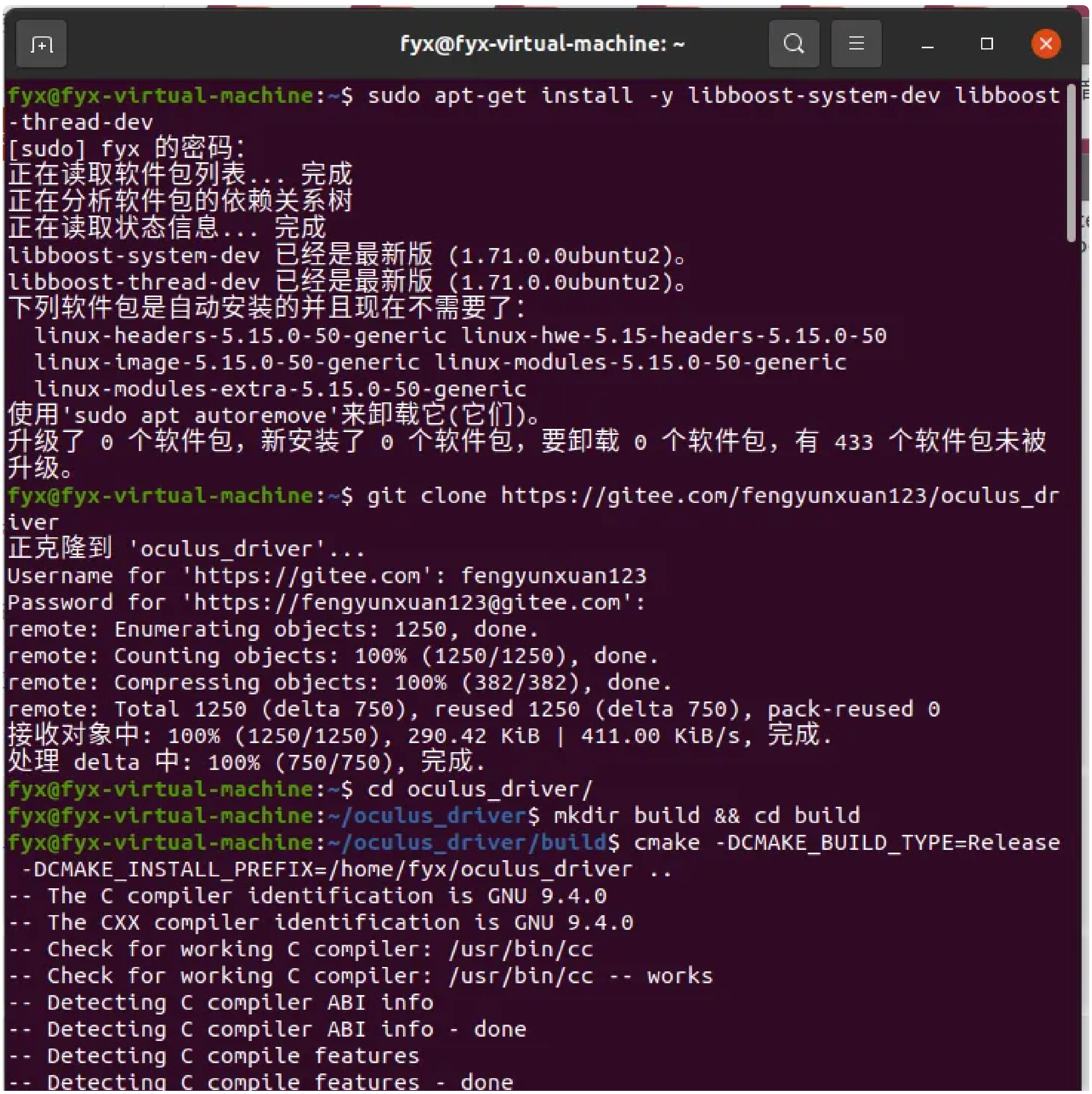
# Bluerov硬件升级&环境配置&使用说明

## 1.配置小主机Intel NUC

装ubuntu20.04系统，安装ROS（安装教程）

## 2.安装声呐驱动&声呐使用

1. 首先安装oculus_driver：，它的用处是在没有网络连接的情况下也可以使用oculus_sonar。按照链接中的readme安装即可



```
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Found Boost: /usr/lib/x86_64-linux-gnu/cmake/Boost-1.71.0/BoostConfig.cmake
(found version "1.71.0") found components: system thread
-- Configuring done
-- Generating done
-- Build files have been written to: /home/fyx/oculus_driver/build
fyx@fyx-virtual-machine:~/oculus_driver/build$ make -j4 install
Scanning dependencies of target oculus_driver
[ 42%] Building CXX object CMakeFiles/oculus_driver.dir/src/SonarClient.cpp.o
[ 42%] Building CXX object CMakeFiles/oculus_driver.dir/src/StatusListener.cpp.
o
[ 57%] Building CXX object CMakeFiles/oculus_driver.dir/src/SonarDriver.cpp.o
[ 57%] Building CXX object CMakeFiles/oculus_driver.dir/src/print_utils.cpp.o
[ 71%] Building CXX object CMakeFiles/oculus_driver.dir/src/AsyncService.cpp.o
[ 85%] Building CXX object CMakeFiles/oculus_driver.dir/src/Recorder.cpp.o
[100%] Linking CXX shared library liboculus_driver.so
[100%] Built target oculus_driver
Install the project...
-- Install configuration: "Release"
-- Installing: /home/fyx/oculus_driver/lib/liboculus_driver.so
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverTar
gets.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverTar
gets-release.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverCon
fig.cmake
-- Installing: /home/fyx/oculus_driver/lib/cmake/oculus_driver/oculus_driverCon
figVersion.cmake
```

在./bashrc文件中添加：

```
2 export CMAKE_PREFIX_PATH=/home/fyx/oculus_driver:$CMAKE_PREFIX_PATH
```

```
fyx@NUC11PAHi5:~$ source ~/.bashrc
fyx@NUC11PAHi5:~$ echo $CMAKE_PREFIX_PATH
/home/fyx/oculus_driver:/opt/ros/noetic:/home/fyx/oculus_driver
```

图中代码根据安装位置适当更换

2.安装oculus_ros功能包

oculus_sonar功能包

创建好一个工作空间，把上面链接里面的文件下载到src文件夹里面解压好，返回工作空间catkin_make

运行launch文件的代码： `roslaunch oculus_sonar sonar_oculus_ros.launch`

launch文件里面有一个路径更改：

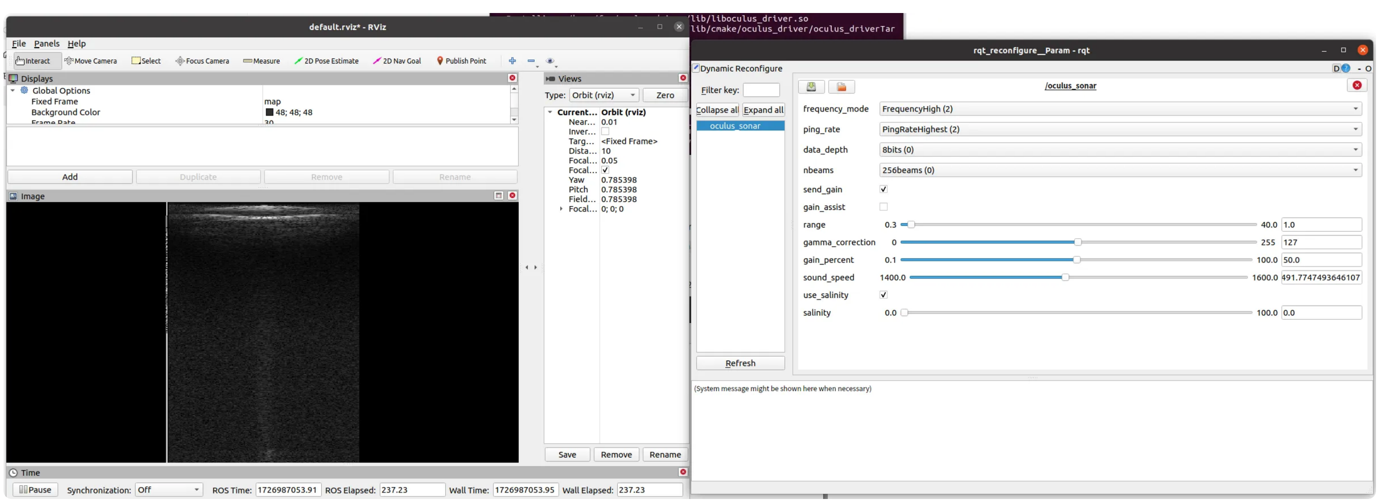
```
    <!-- 打开rviz -->
    <node pkg="rviz" type="rviz" name="rviz" args="-d ~/oculus_ros/src/sonar_oculus/cfg/-
sonar_rviz.rviz" required="true" />
```

oculus_viewer.py允许文件作为程序执行打钩

使用：

插上网线后，有线连接的IVP4地址改为：~~169.254.70.29（也可能是169.254.179.91）~~


调整参数后就可以在rviz中看到图像：




# 3.安装XSENS MTI300 IMU驱动

XSENS驱动软件包下载，解压

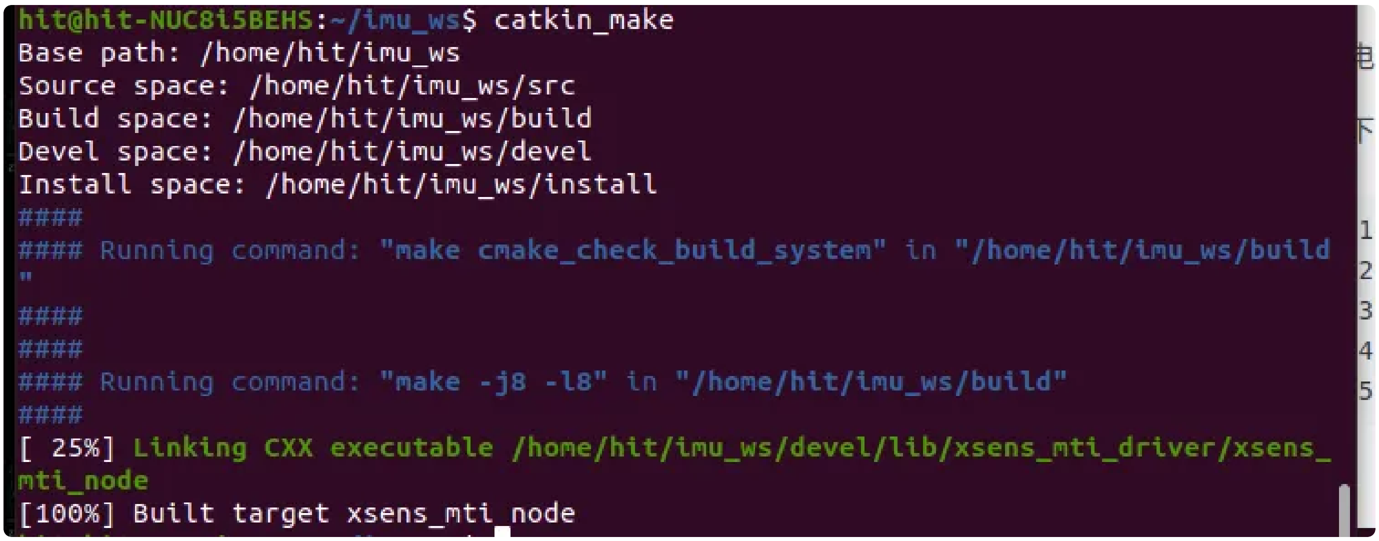
打开终端， `sudo apt-get install sharutils`

~/下载/MT_Software_Suite_linux-x64_2020.5路径下， `sudo ./mtsdk_linux-x64_2020.5.sh`

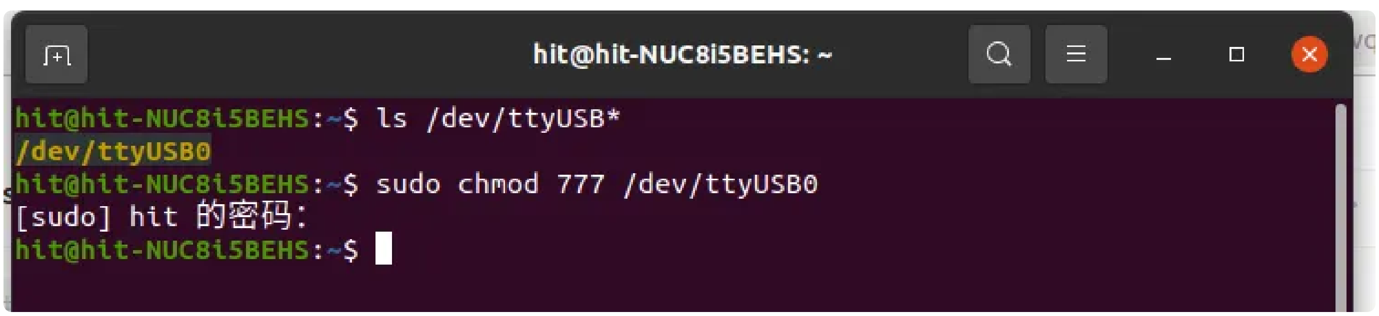
创建好一个工作空间imu_ws，将/usr/local/xsens文件夹中的xsens_ros_mti_driver文件夹复制到工作空间src下

~/imu_ws/src/xsens_ros_mti_driver/lib/xspublic路径下， `make`

返回工作空间目录， `catkin_make`

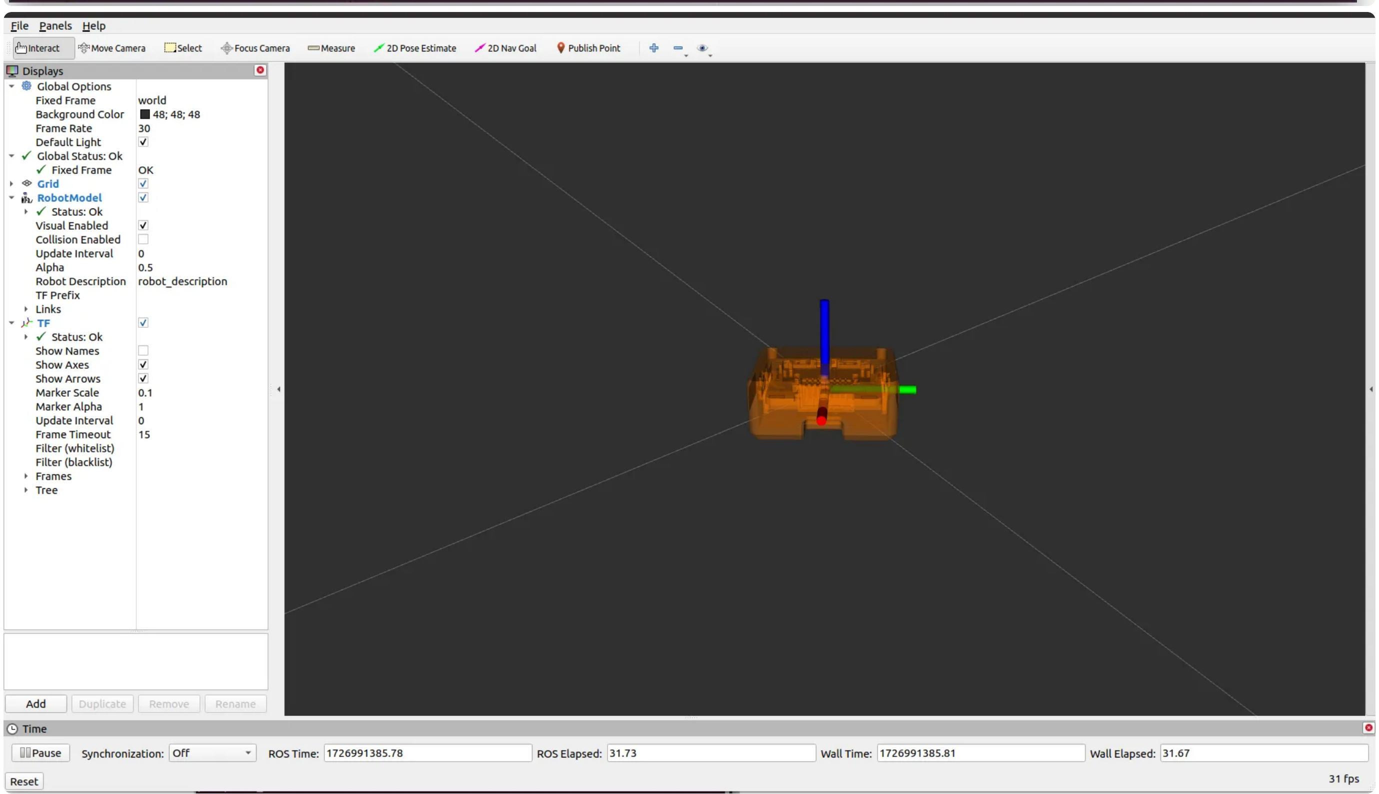



imu连接设备，查看串口编号，并给权限：




`roslaunch xsens_mti_driver display.launch` 就可以启动设备

```
hit@hit-NUC8i5BEHS:~$ rostopic list
/clicked_point
/filter/free_acceleration
/filter/positionlla
/filter/quaternion
/filter/twist
/filter/velocity
/gnss
/imu/acceleration
/imu/angular_velocity
/imu/data
/imu/dq
/imu/dv
/imu/mag
/imu/time_ref
/initialpose
/move_base_simple/goal
/pressure
/rosout
/rosout_agg
/temperature
/tf
/tf_static
hit@hit-NUC8i5BEHS:~$
```

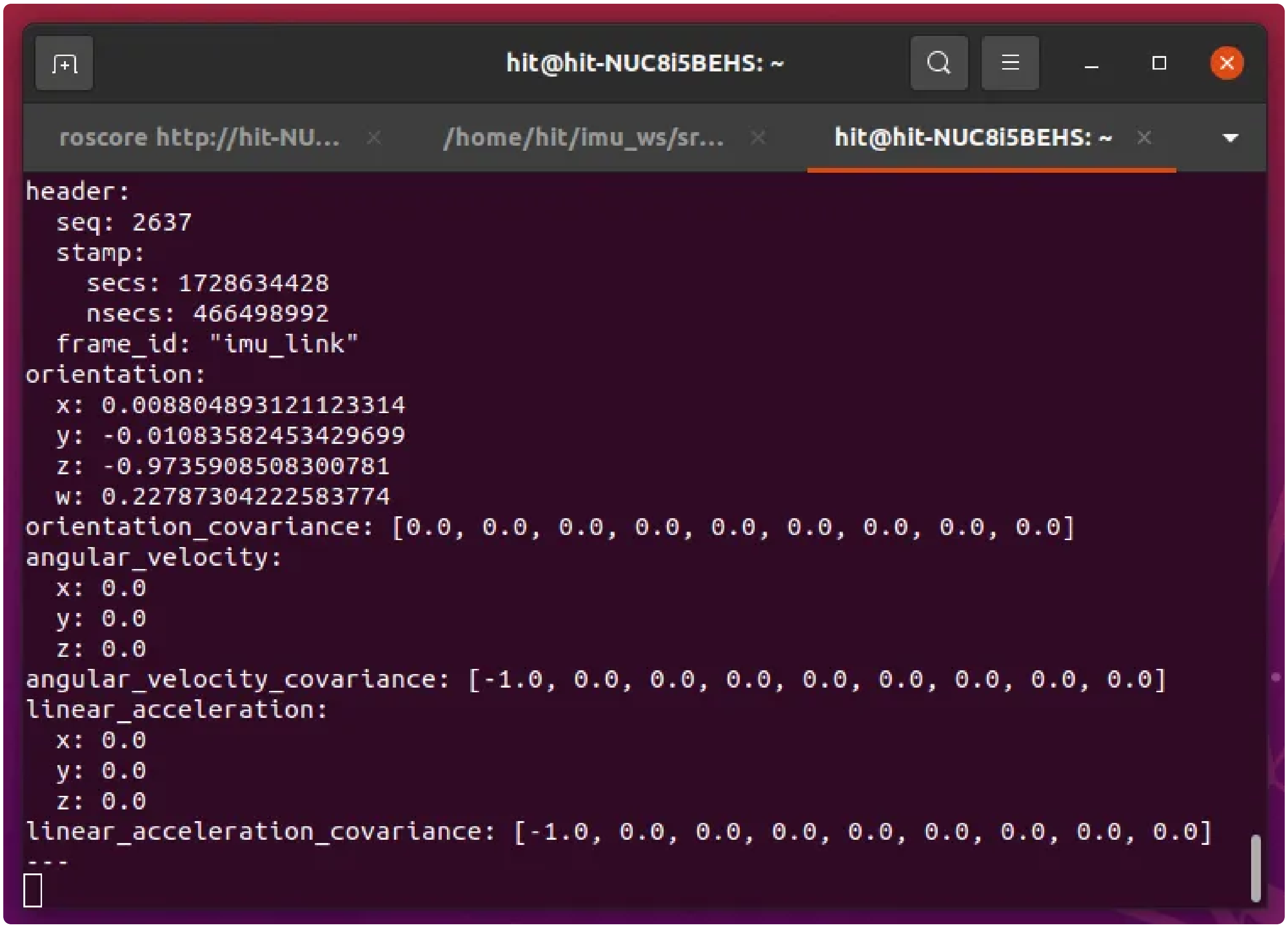



现在每次开机插入设备都需要重新给设备权限，sudo chmod 777 /dev/ttyUSB0

解决方法：

`sudo usermod -aG dialout $USER（用户名）`，重启电脑

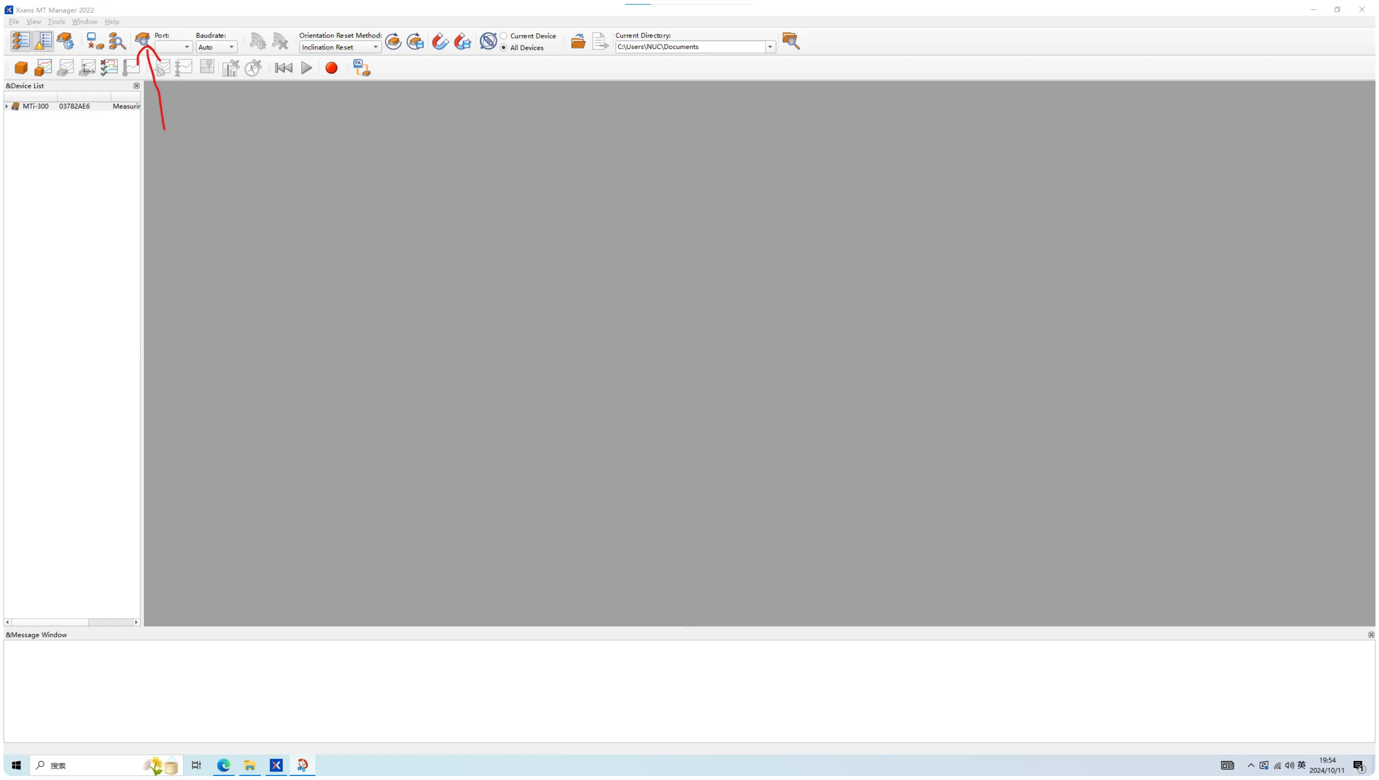
ROS输出没有角速度和线加速度信息：



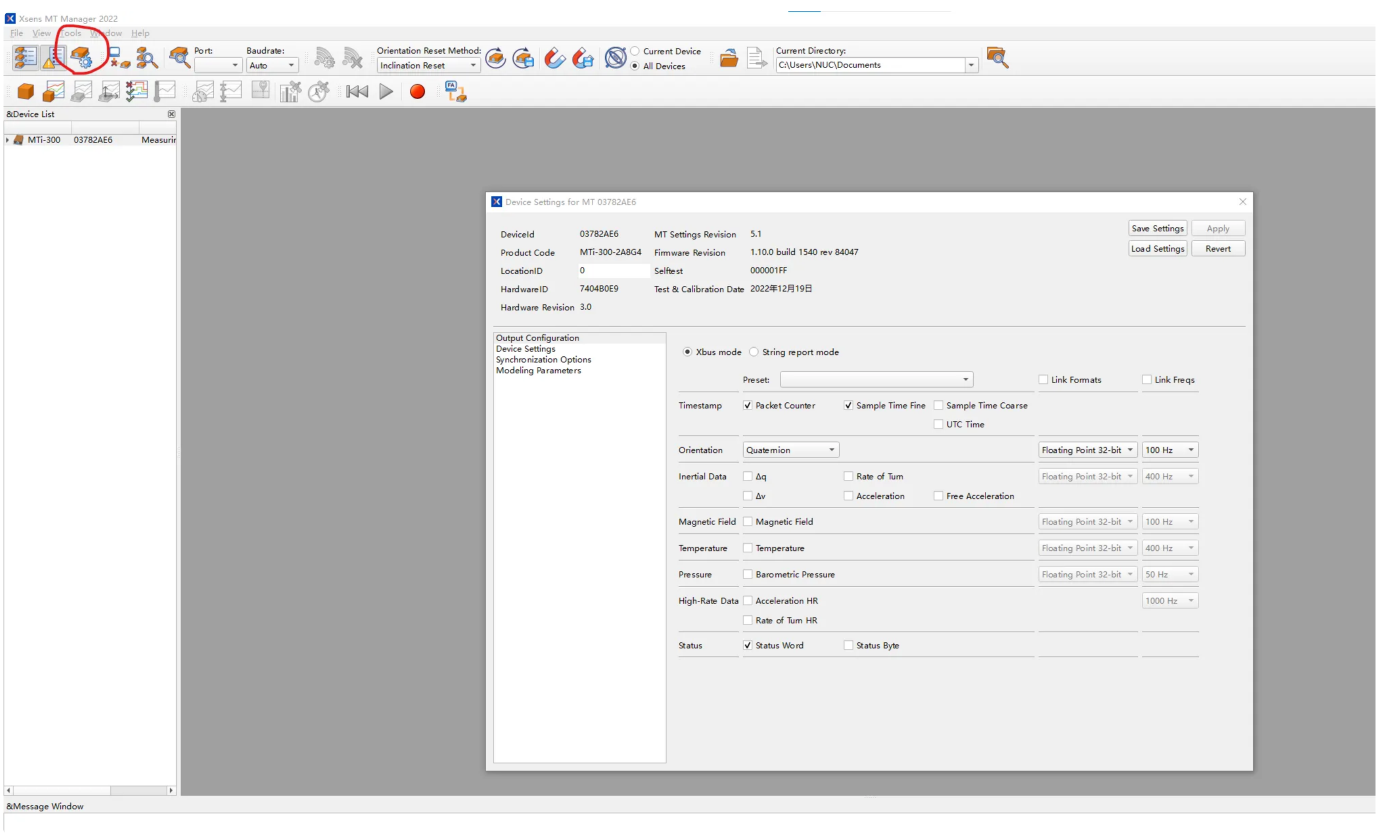
需要使用MT Manager来更改输出配置

MT Manager下载网站，下载windows版本就可以

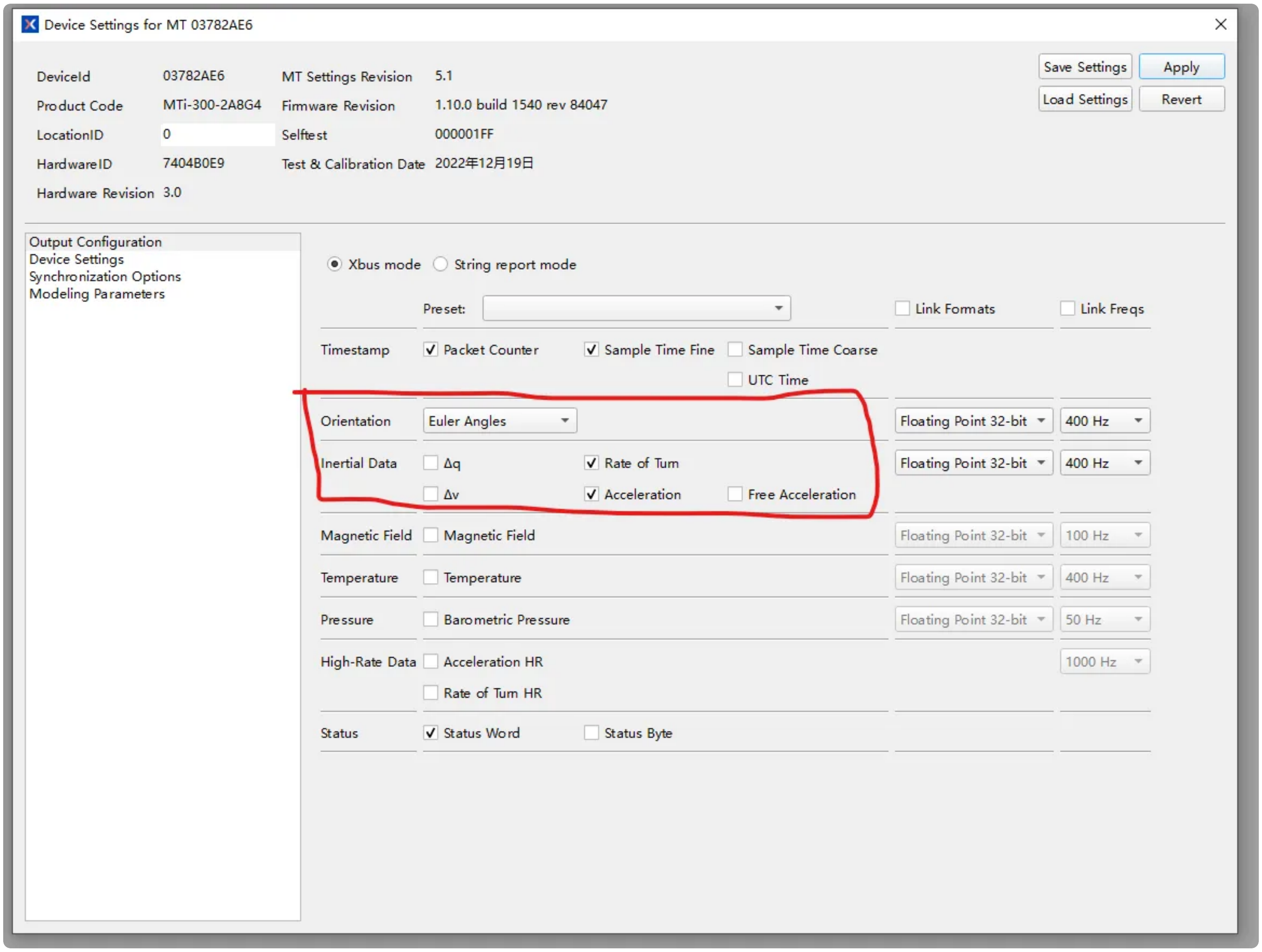
打开界面，连接IMU，点击扫描端口按钮，扫描设备



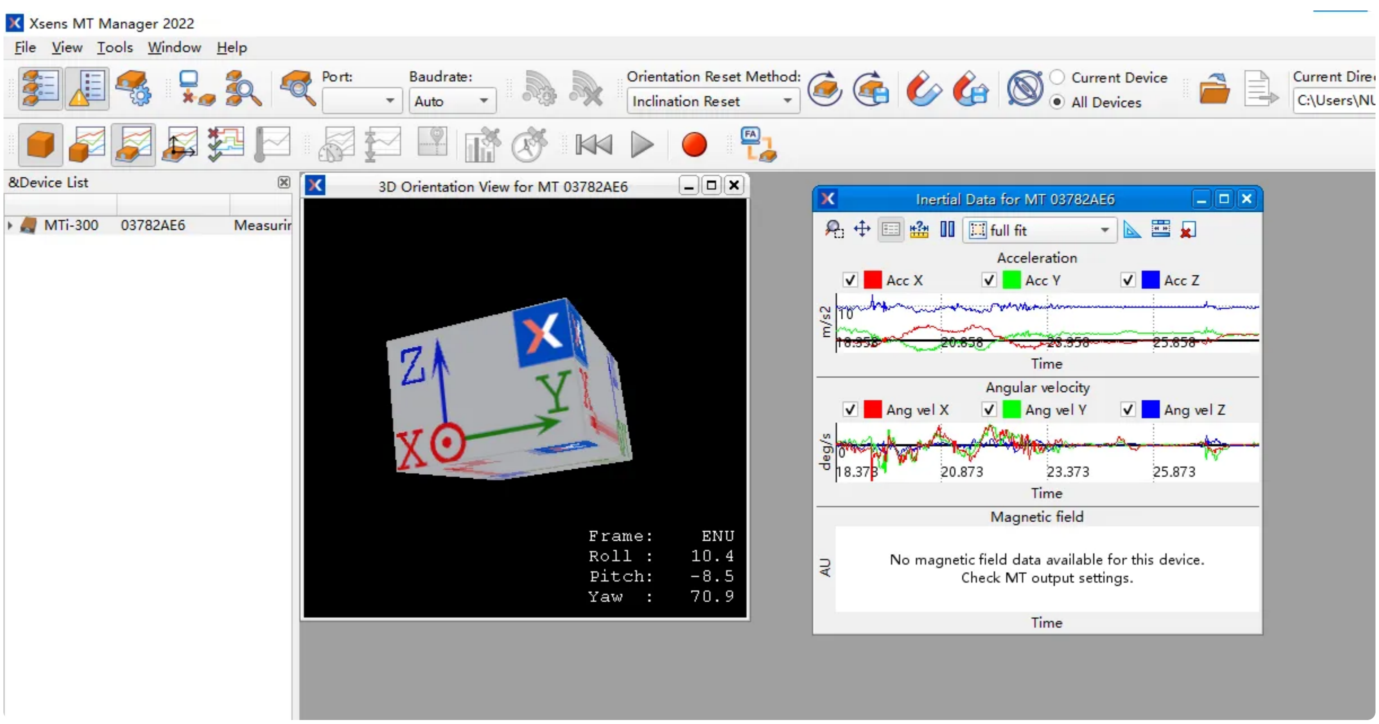

左边选中设备，点击配置参数按钮



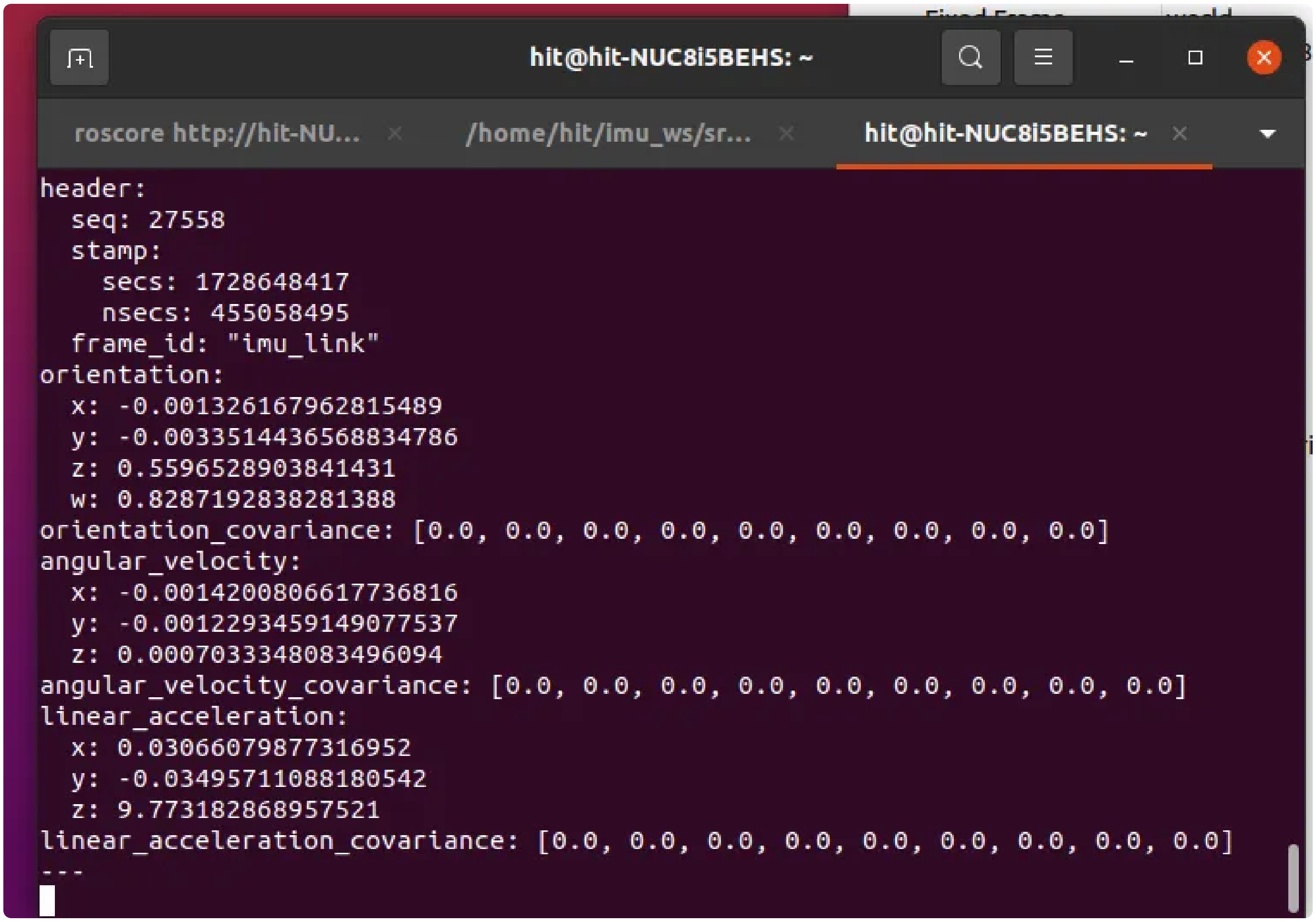

选择角速度和加速度，右上角应用




可以看见姿态显示与角速度、加速度输出




回到ubuntu ros中也有输出了：




## 4.安装BlueRov驱动与ROS功能包

bluerov_ros_playground

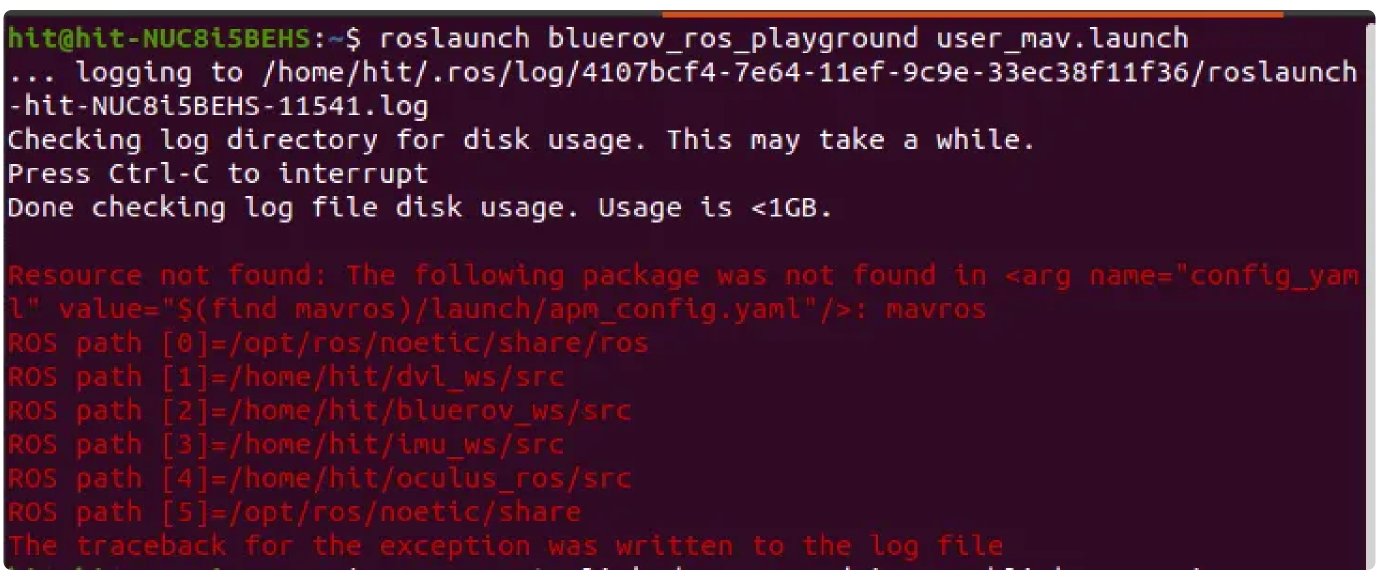
创建好一个工作空间bluerov_ws，bluerov_ws/src文件夹下，克隆上面的功能包，返回工作空间catkin_make

安装QGC

USB以太网IVP4设置：192.168.2.1


运行： `roslaunch bluerov_ros_playground user_mav.launch`

出现以下错误是因为没有安装mavros包，按照mavros安装即可

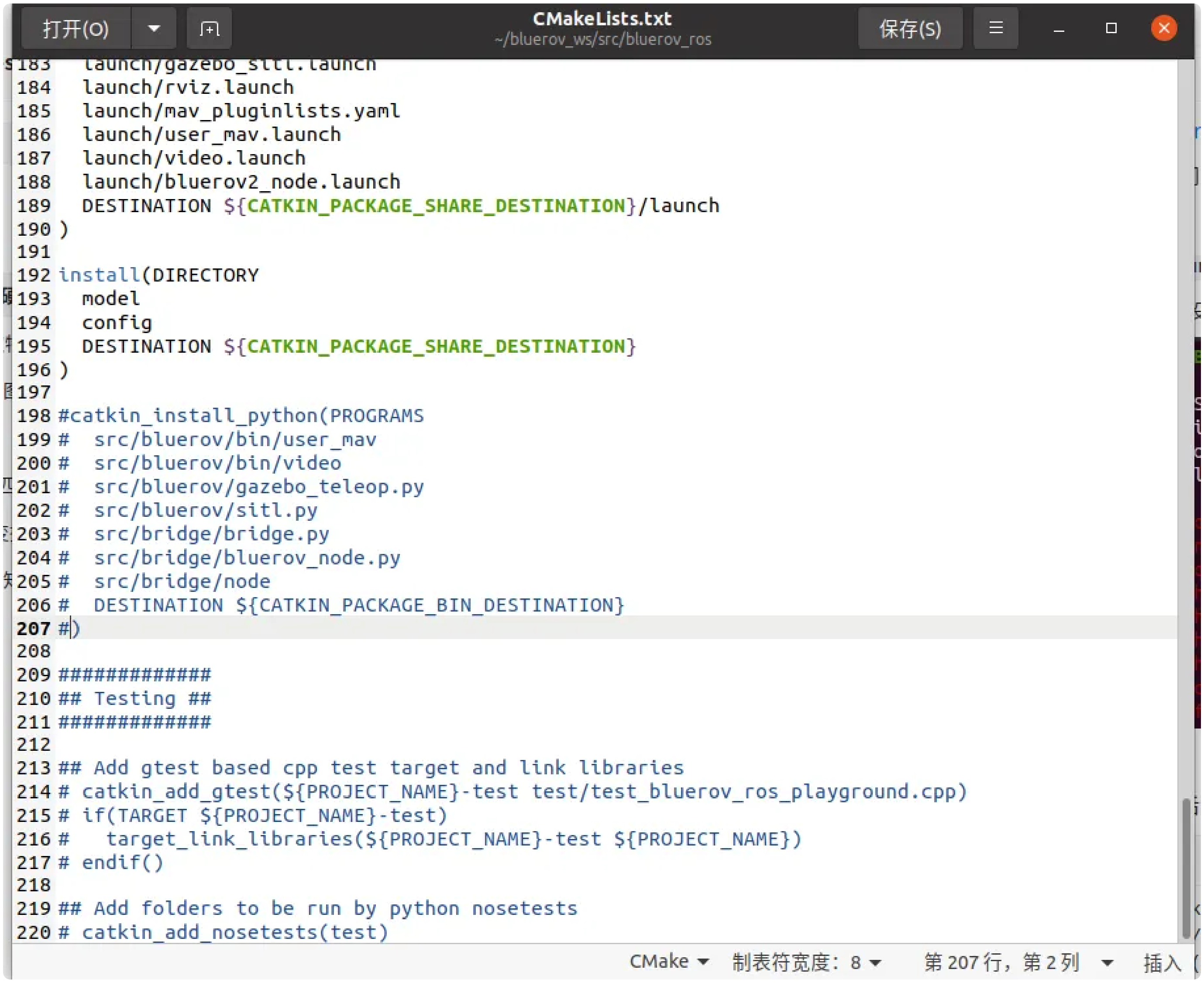



上面安装的包运行后会报错：

```
Traceback (most recent call last):
  File "/home/hit/bluerov_ws/devel/lib/bluerov_ros_playground/node", line 15, in <module>
    exec(compile(fh.read(), python_script, 'exec'), context)
  File "/home/hit/bluerov_ws/src/bluerov_ros/src/bridge/node", line 4, in <module>
    import bluerov_node as BR
  File "/home/hit/bluerov_ws/devel/lib/bluerov_ros_playground/bluerov_node.py", line 15, in <module>
    exec(compile(fh.read(), python_script, 'exec'), context)
  File "/home/hit/bluerov_ws/src/bluerov_ros/src/bridge/bluerov_node.py", line 12, in <module>
    from bridge import Bridge
ImportError: cannot import name 'Bridge' from 'bridge' (/home/hit/bluerov_ws/devel/lib/bluerov_ros_playground/bridge.py)
```

解决办法：

/home/hit/bluerov_ws/src/bluerov_ros/CMakeList.txt文件198-207行注释掉：




```
  launch/gazebo_sitl.launch
  launch/rviz.launch
  launch/mav_pluginlists.yaml
  launch/user_mav.launch
  launch/video.launch
  launch/bluerov2_node.launch
  DESTINATION ${CATKIN_PACKAGE_SHARE_DESTINATION}/launch
)

install(DIRECTORY
  model
  config
  DESTINATION ${CATKIN_PACKAGE_SHARE_DESTINATION}
)

#catkin_install_python(PROGRAMS
#   src/bluerov/bin/user_mav
#   src/bluerov/bin/video
#   src/bluerov/gazebo_teleop.py
#   src/bluerov/sitl.py
#   src/bridge/bridge.py
#   src/bridge/bluerov_node.py
#   src/bridge/node
#   DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
#)

#############
## Testing ##
#############

## Add gtest based cpp test target and link libraries
# catkin_add_gtest(${PROJECT_NAME}-test test/test_bluerov_ros_playground.cpp)
# if(TARGET ${PROJECT_NAME}-test)
#   target_link_libraries(${PROJECT_NAME}-test ${PROJECT_NAME})
# endif()

## Add folders to be run by python nosetests
# catkin_add_nosetests(test)
```

另外，将所有.py文件的第一行由 `#!/usr/bin/env python` 改为 `#!/usr/bin/env python3`

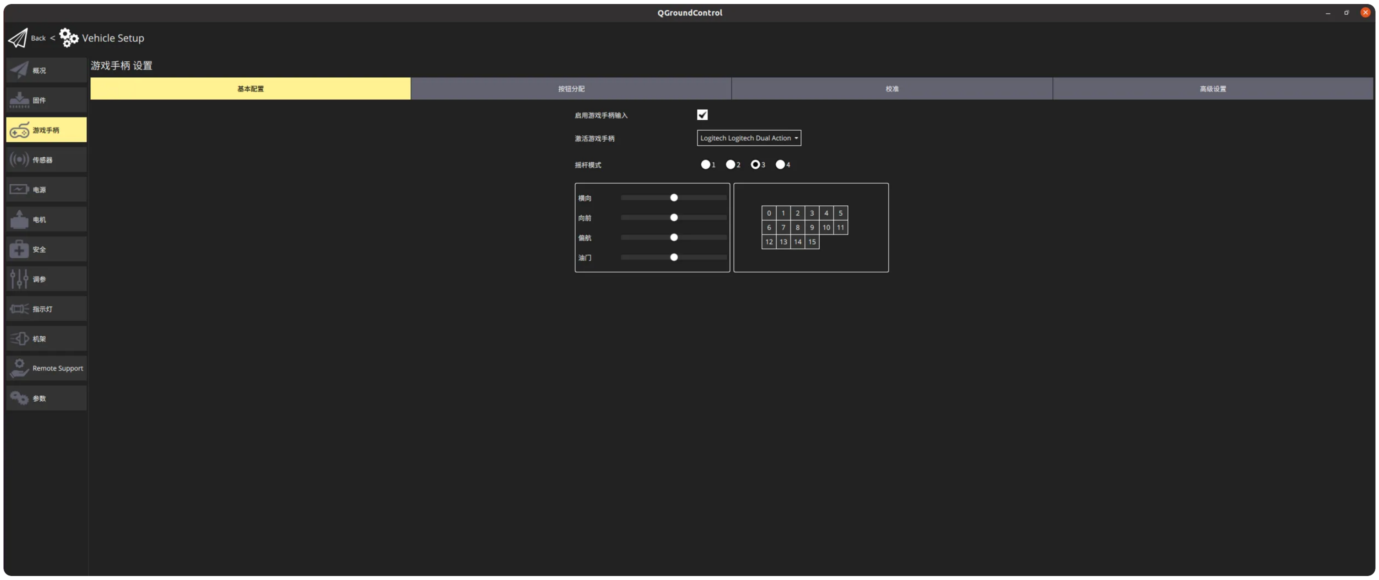
更改后，运行 `roslaunch bluerov_ros_playground user_mav.launch` 可以正常运行，输出数据并弹出视觉摄像头窗口：


`rostopic echo /mavros/imu/data` 输出（机器人自带imu数据）：

```
---
header:
  seq: 1811
  stamp:
    secs: 1727621179
    nsecs:  25567073
  frame_id: "base_link"
orientation:
  x: -0.05530324325263004
  y: 0.017811554402223567
  z: 0.20422853198608543
  w: -0.9771975268789446
orientation_covariance: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]
angular_velocity:
  x: -0.0005168463103473186
  y: 1.123920083045962e-05
  z: 0.00020210817456245422
angular_velocity_covariance: [1.2184696791468346e-07, 0.0, 0.0, 0.0, 1.218469679
1468346e-07, 0.0, 0.0, 0.0, 1.2184696791468346e-07]
linear_acceleration:
  x: 0.12748645
  y: 1.5984839500000012
  z: 10.1204628
linear_acceleration_covariance: [8.999999999999999e-08, 0.0, 0.0, 0.0, 8.9999999
```

~~QGC里解锁以后动手柄机器人螺旋桨没反应，不知道是哪里的问题~~

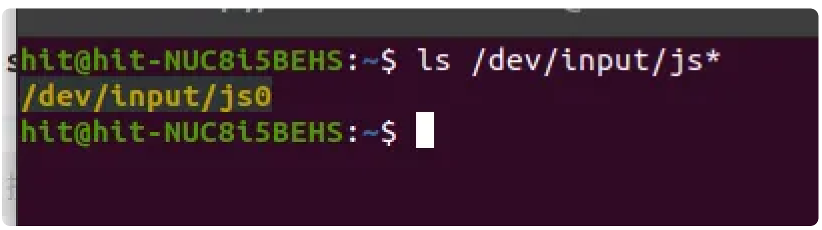
QGC里设置Vehicle Setup里把启动游戏手柄输入的钩钩上




有关检查手柄：

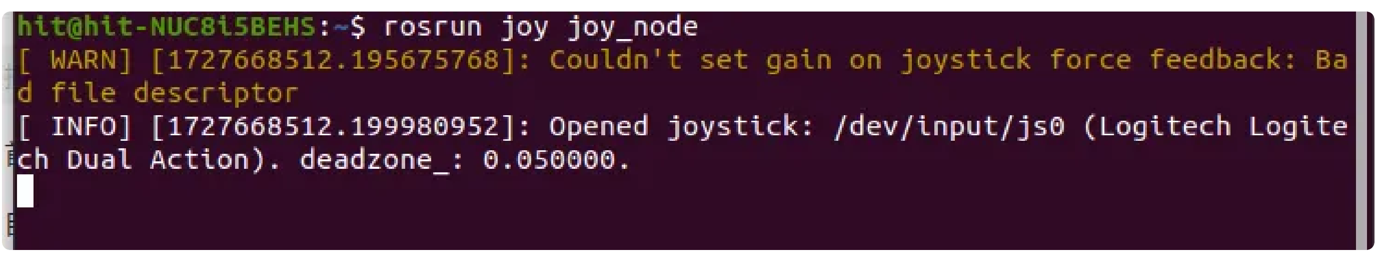
终端里：

`ls /dev/input/js*` 检查设备：

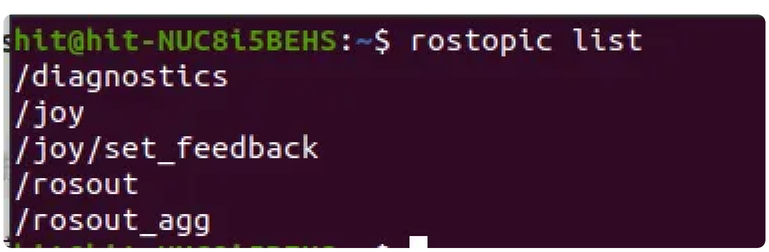



`sudo chmod 777 /dev/input/js0` 给权限

`rosrun joy joy_node` 启动手柄ros节点：（如图则为检测到手柄信息）




检查话题：



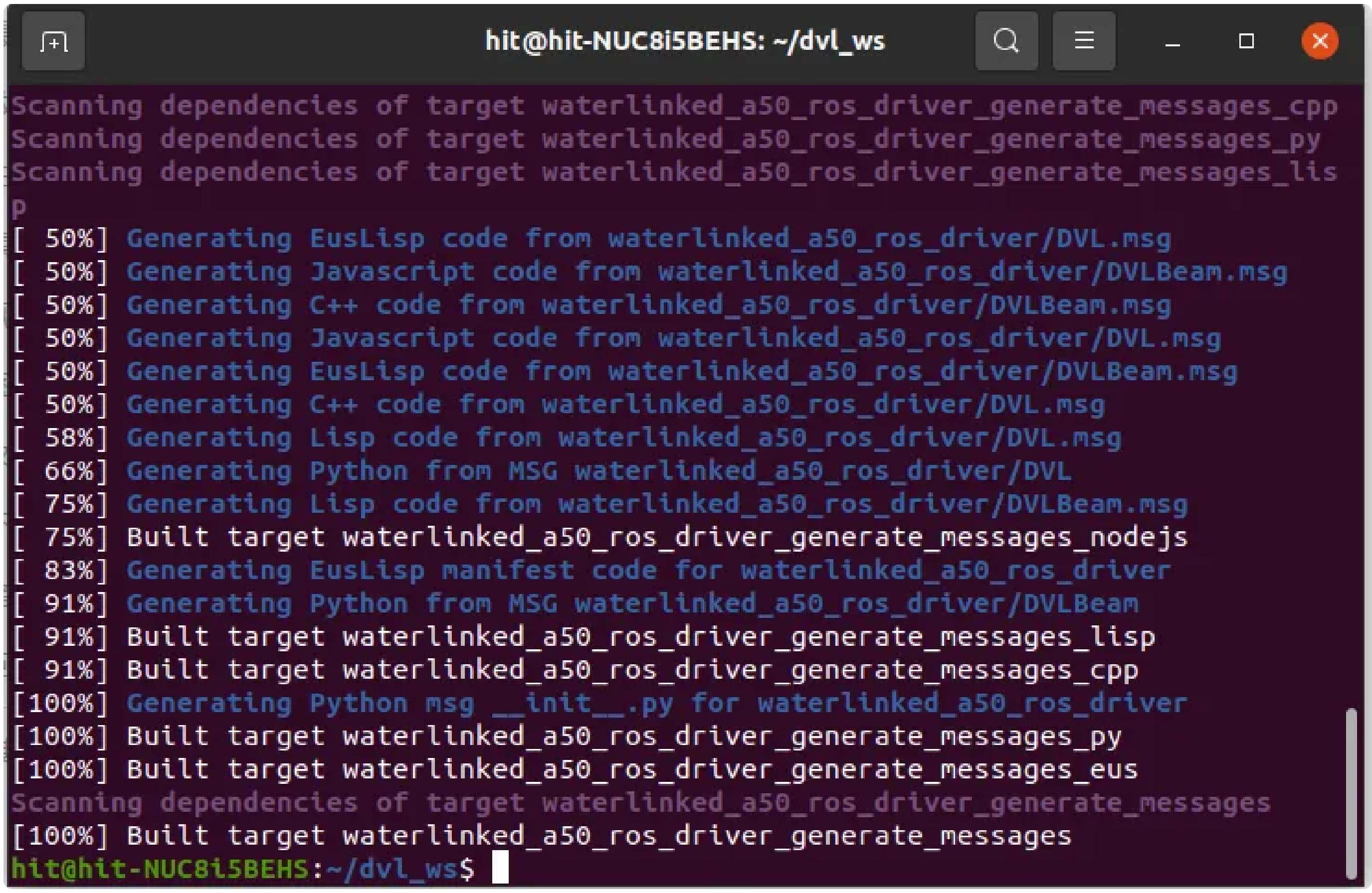
```
hit@hit-NUC8i5BEHS:~$ rostopic echo /joy
header:
  seq: 1
  stamp:
    secs: 1727668622
    nsecs: 984244120
  frame_id: "/dev/input/js0"
axes: [0.0, 0.0, -0.08842818439006805, 0.0, 0.0, 0.0]
buttons: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
---
header:
  seq: 2
  stamp:
    secs: 1727668622
    nsecs: 992205294
  frame_id: "/dev/input/js0"
axes: [0.0, 0.0, -0.2620624303817749, 0.0, 0.0, 0.0]
buttons: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

正确输出包含axes和buttons的字段信息则说明手柄没有问题

## 5.安装DVL驱动与使用

DVL A50

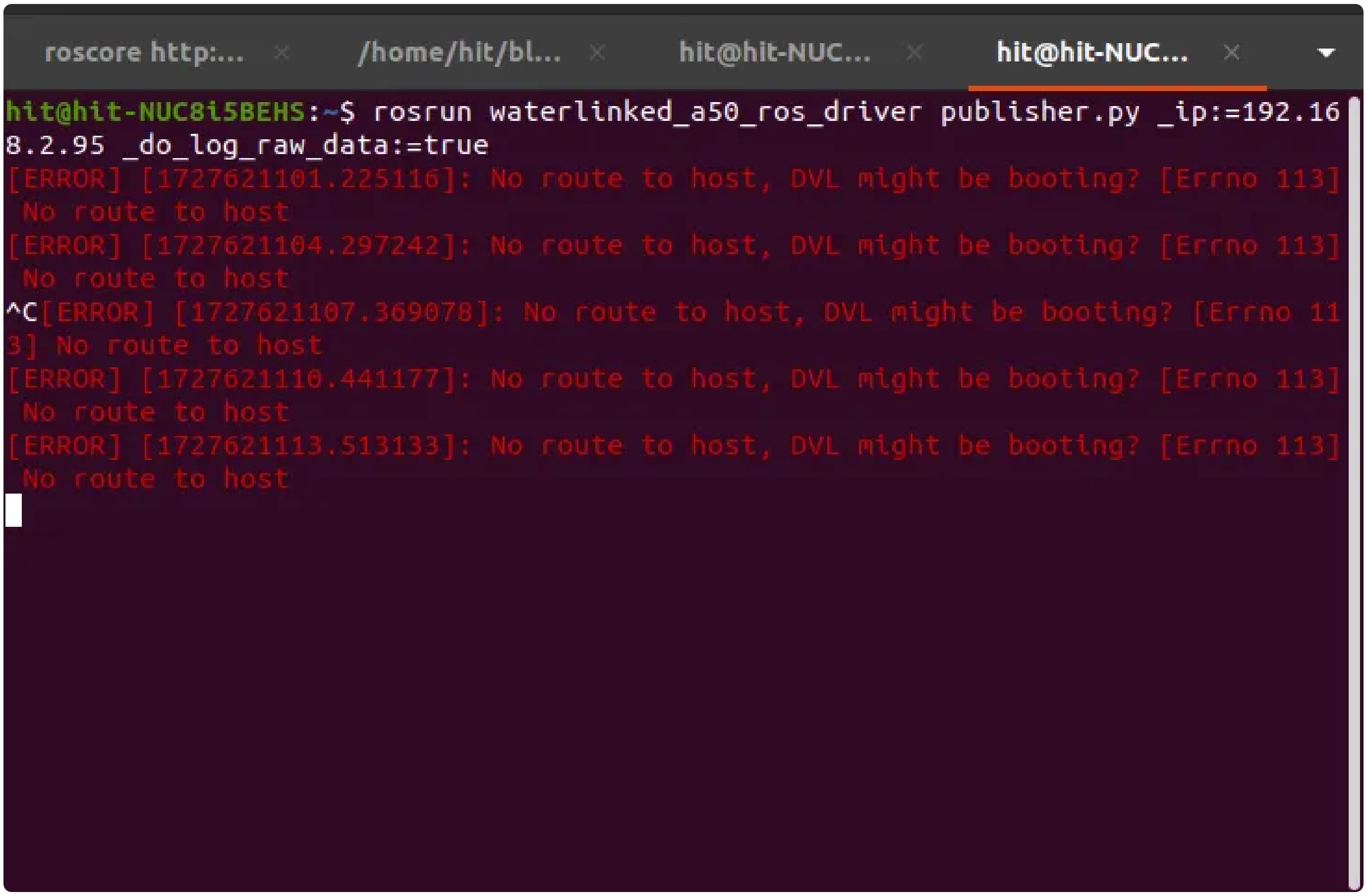
创建好一个工作空间dvl_ws， dvl_ws/src文件夹下，克隆上面的功能包，返回工作空间catkin_make




运行：`rosrun waterlinked_a50_ros_driver publisher.py _ip:=192.168.194.95 _do_log_raw_data:=true`

报错：[ERROR] [1727669726.186719]: No route to host, DVL might be booting? [Errno 111] Connection refused

~~待解决~~是因为ip的问题




小主机的ip设置为：192.168.194.90


运行 `rosrun waterlinked_a50_ros_driver publisher.py _ip:=192.168.194.95 _do_log_raw_data:=true` 后报错： `TypeError: can only concatenate str (not "bytes") to str`

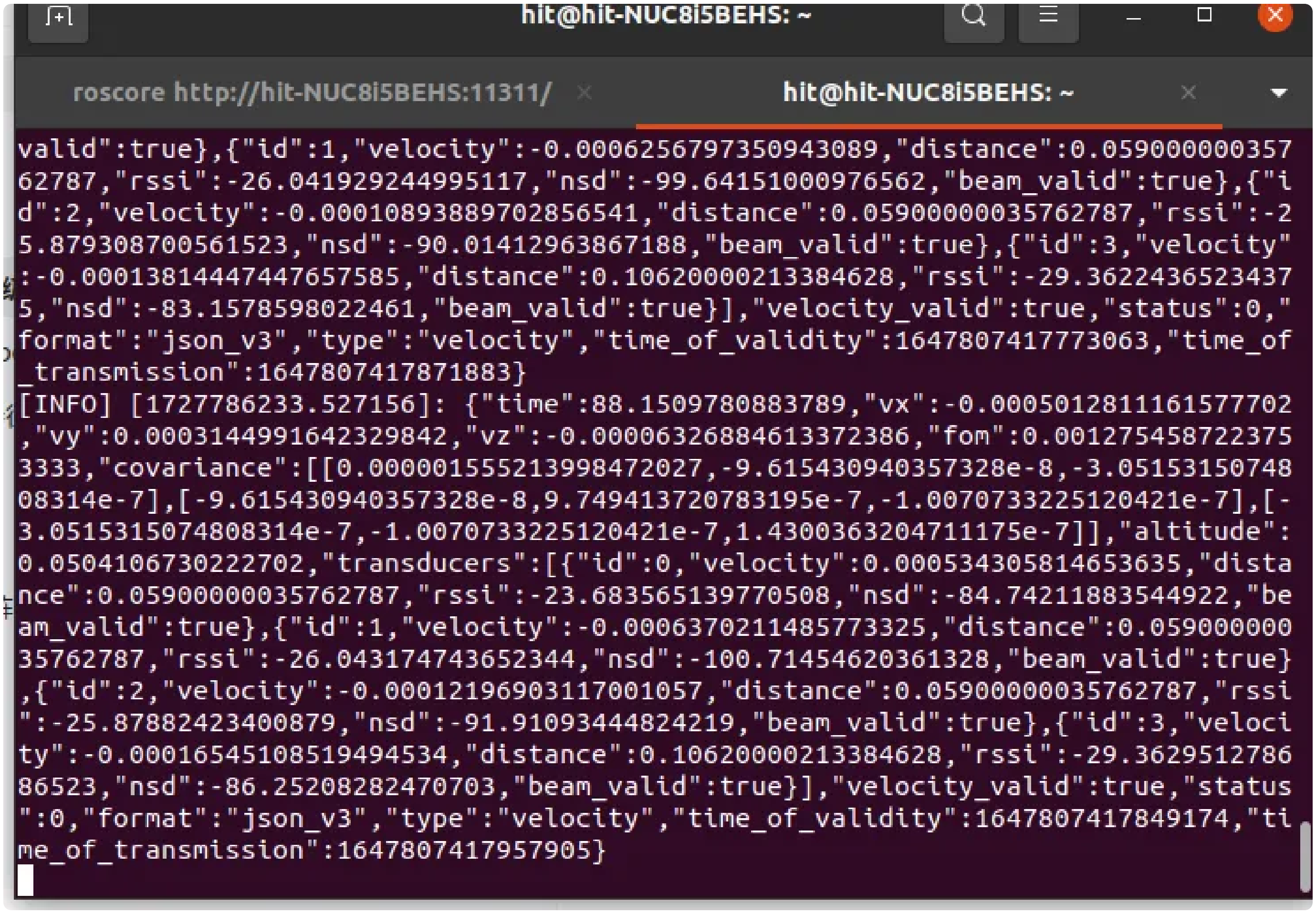
根据信息找到错误行： `raw_data = raw_data + rec` 试图将字节（bytes）类型与字符串（str）类型拼接，这在Python3中是不允许的

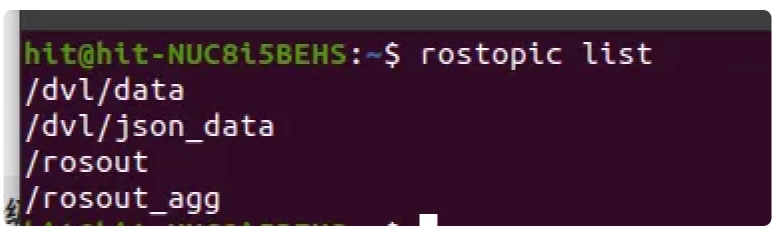
更改为： `raw_data = raw_data + rec.decode('utf-8') # 将bytes转换为str`

```
def getData():
	global oldJson, s
	raw_data = ""

	while not '\n' in raw_data:
		try:
			rec = s.recv(1) # Add timeout for that
			if len(rec) == 0:
				rospy.logerr("Socket closed by the DVL, reopening")
				connect()
				continue
		except socket.timeout as err:
			rospy.logerr("Lost connection with the DVL, reinitiating the connection: {}".format(err))
			connect()
			continue
		raw_data = raw_data + rec.decode('utf-8') # 将bytes转换为str
	raw_data = oldJson + raw_data
	oldJson = ""
	raw_data = raw_data.split('\n')
	oldJson = raw_data[1]
	raw_data = raw_data[0]
	return raw_data
```

修改后运行指令可以看到DVL数据输出：




# 6.下载nomachine远程桌面软件

nomachine下载

安装教程按照教程将压缩包复制进/usr后操作

```
fyx@NUC11PAHi5:~$ su
密码:
root@NUC11PAHi5:/home/fyx# cd 下载
root@NUC11PAHi5:/home/fyx/下载# cp -r nomachine_8.14.2_1_x86_64.tar.gz /usr
root@NUC11PAHi5:/home/fyx/下载# cd
root@NUC11PAHi5:~# cd /usr
root@NUC11PAHi5:/usr# sudo tar zxvf nomachine_8.14.2_1_x86_64.tar.gz
NX/
NX/nxserver
NX/etc/
NX/etc/NX/
NX/etc/NX/server/
NX/etc/NX/server/localhost/
NX/etc/NX/server/localhost/server.cfg.sample
NX/etc/NX/server/packages/
NX/etc/NX/server/packages/nxnode.tar.gz
NX/etc/NX/server/packages/nxrunner.tar.gz
NX/etc/NX/server/packages/nxplayer.tar.gz
NX/etc/NX/server/packages/nxserver.tar.gz
README-NOMACHINE
root@NUC11PAHi5:/usr# sudo /usr/NX/nxserver --install
NX> 700 Starting installation at: Thu, 31 Oct 2024 17:23:39.
NX> 700 Using installation profile: Ubuntu.
NX> 700 Installation log is: /usr/NX/var/log/nxinstall.log.
NX> 700 Installing nxrunner version: 8.14.2.
NX> 700 Installing nxplayer version: 8.14.2.
NX> 700 Installing nxnode version: 8.14.2.
NX> 700 Installing nxserver version: 8.14.2.
NX> 700 Installation completed at: Thu, 31 Oct 2024 17:23:50.
NX> 700 NoMachine was configured to run the following services:
NX> 700 NX service on port: 4000
root@NUC11PAHi5:/usr#
```

小主机拔掉HDMI线后系统会自动关闭显示输出，因此远程桌面会黑屏，解决办法：

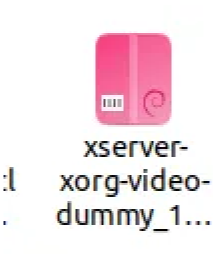
（因为已经装进密封舱连好线了，因此尝试以下不需要联网的方法）

1）在另一台联网的电脑上下载.deb包

```
sudo apt update
apt download xserver-xorg-video-dummy
```

```
hit@hit-NUC8i5BEHS:~$ apt download xserver-xorg-video-dummy
获取:1 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal/main amd64 xserver-xorg
-video-dummy amd64 1:0.3.8-1build3 [8,884 B]
已下载 8,884 B，耗时 0秒 (26.8 kB/s)
```

这会在主目录下载.deb安装包，用U盘将此文件拷入小主机




2）小主机中，

`sudo dpkg -i  xserver-xorg-video-dummy_*.deb`

3）在文件夹 /usr/share/X11/xorg.conf.d中，创建xorg.conf文件，写入以下内容：

Plain Text

```
Section "Device"
    Identifier "Configured Video Device"
    Driver "dummy"
    VideoRam 256000
EndSection

Section "Monitor"
    Identifier "Configured Monitor"
    HorizSync 28.0-80.0
    VertRefresh 48.0-75.0
    Modeline "1920x1080_60.00" 172.80 1920 2040 2248 2576 1080 1081 1084 1118 -HSync +Vsync
EndSection

Section "Screen"
    Identifier "Default Screen"
    Monitor "Configured Monitor"
    Device "Configured Video Device"
    DefaultDepth 24
    SubSection "Display"
        Depth 24
        Modes "1920x1080_60.00"
    EndSubSection
EndSection
```

4）保存，重启系统

此时拔掉HDMI线，连接nomachine就可以控制了

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# 7.动捕配置及使用

需要下载相关的功能包：

3. VRPN 客户端的下载及网络配置，使用虚拟机运行（如图 9-11）
    - cd ~/catkin_ws/src
    - git clone https://github.com/clearpathrobotics/vrpn_client_ros.git
    - sudo apt-get install ros-kinetic-vrpn

使用说明📎Seeker与ROS的通信.pdf